



BESIII Analysis Memo

BAM-xxx

BESIII Analysis Memo (January 5, 2026)

Cross section measurement of process $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

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20 **List of changes**

21 For an easy reading of the document, major changes in various versions with respect to the previous
22 one are listed below.

23 **Version 1.0**

- 24 • First version of memo.

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1 Introduction

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2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider—is operating in the τ -charm energy region (2 to 4.9 GeV) with the design luminosity of $1 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$ [1]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [2]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [3].

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1} , collected by BESIII detector at center-of-mass energy of 3.773 GeV , under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

$E_{\text{beam}}(\text{GeV})$	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	$2931.8 \pm 0.2 \pm 13.8$ [4]	2010(round03) 2011(round04)	11414 – 13988, 14395 – 14604 20448 – 23454
3.773	4995.0 ± 19.0 [5]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [5]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [5]	2024(round17)	78615 – 81094

Simulated Monte Carlo (MC) samples are produced with the GEANT4-based [6] MC package. Two types of MC samples are used in this analysis as described in the following:

- Inclusive MC samples, which includes the production of $\Lambda_c^+ \bar{\Lambda}_c^-$ pairs, open-charmed mesons, ISR production of vector charmonium(-like) states, continuum processes incorporated in KKMC [?, ?]. Known decay modes are modeled with EVTGEN [?, ?] using the BFs taken from the PDG [?]. The remaining unknown charmonium decays are modeled with LUNDCHARM [?, ?]. The integrated luminosity is 40 times larger than that of data, and the $e^+e^- \rightarrow \Lambda_c^+ \bar{\Lambda}_c^-$ cross section is based on the analysis of Ref. [?]. For $\Lambda_c^+ \bar{\Lambda}_c^-$ part (denoted as inclusive $\Lambda_c^+ \bar{\Lambda}_c^-$ MC samples), the decay card and the dst files are available in the path /besfs5/offline/data/706-1/xyz/mc/*/LcLc2. For non- $\Lambda_c^+ \bar{\Lambda}_c^-$ part (denoted as hadron MC samples), the samples are available in the path /besfs5/offline/data/706-1/xyz/mc/*/hadron. See Ref. [?, ?] for more details about the generation of official MC samples.
- PHSP signal MC samples for the process $\Lambda_c^+ \rightarrow p K_S^0 \pi^0$ (charge-conjugated mode implied) are generated in two configurations:
 - **Inclusive $\bar{\Lambda}_c^-$ decay samples** modeling all possible decay channels. These samples are used in the MC integration in the likelihood construction of the amplitude analysis fit where the detection efficiencies are therefore implemented. At each center-of-mass energy point 500K signal MC events are generated.
 - **Detector-invisible $\bar{\Lambda}_c^-$ decay samples** where final states are simulated as undetectable. This configuration yields a pure signal sample employed for training the Boosted Decision Tree with Gradient Boosting (BDTG) classifier.

⁷⁴ **4 Event selections**

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5 Summary

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References

- [1] M. Ablikim *et al.* (BESIII Collaboration), *Future Physics Programme of BESIII*, *Chinese Phys. C* **44** (2020).
- [2] R. X. Yang *et al.*, *MRPC detector for the BESIII E-TOF upgrade*, *Journal of Instrumentation*, **Volume 9**, September 2014. DOI: [10.1088/1748-0221/9/09/C09032](https://doi.org/10.1088/1748-0221/9/09/C09032).
- [3] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.nima.2009.12.050](https://doi.org/10.1016/j.nima.2009.12.050).
- [4] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.physletb.2015.11.043](https://doi.org/10.1016/j.physletb.2015.11.043).
- [5] M. Ablikim *et al.* [BESIII Collaboration], DOI: [10.1088/1674-1137/ad70a0](https://doi.org/10.1088/1674-1137/ad70a0).
- [6] S. Agostinelli *et al.* [GEANT4], *Nucl. Instrum. Meth. A* **506**, 250-303 (2003).

87 **APPENDIX**